

Why Are Some ‘Liquid Batteries’ So Thick?

Decoding the Molecular Origins of Viscosity in Ionic and Polymerized Ionic Liquids

A look inside Jester Itliong’s PhD research

01 The Big Picture

Ionic liquids are salts that stay liquid at room temperature, unlike the table salt in your kitchen. These charged liquids, and their polymer cousins, **polymerized ionic liquids**, show real promise for next-generation batteries, carbon capture, and lubricants. The catch is that they are often as thick as syrup, which slows the very ion movement that makes them useful.

ONE FAMILY, THREE FORMS

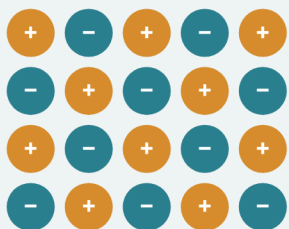
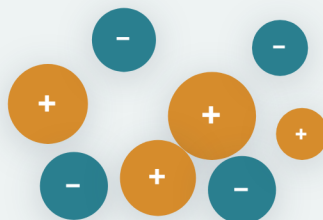
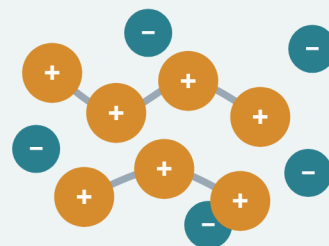


Table salt: solid at room temperature



Ionic liquid: liquid at room temperature but thick



Polymer version: cations tied into chains, thicker than ionic liquid

02 The Question

What molecular features control this thickness? Ion size, electric charge, and chemical structure are all suspects. In real experiments these traits always change together, so isolating a single cause is nearly impossible.

03 The Approach

Using simplified computer simulations, I could dial each molecular feature up or down on its own, something no real experiment allows, and watch exactly how the viscosity responded.

04 The Discovery

KEY FINDING

One factor stood out: how long pairs of oppositely charged ions stick together. Longer-lived pairs make a thicker liquid. Even when ions are tied to long polymer chains, it is this local pairing, not the chain’s wiggling, that mainly sets the thickness.



Ion-pairing is how long opposite charges keep *holding hands*, and that is what controls the flow.

05 Why It Matters

This gives engineers a clear design principle. To build thinner, better-performing electrolytes made of ionic liquids and/or polymerized ionic liquids, **target ion-pairing behavior directly**. Better ionic liquid electrolytes could accelerate next-generation batteries, carbon capture membranes, and industrial lubricants.



Batteries



Carbon Capture



Lubricants

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